

Research Title

Artificial Intelligence in Sports Management

Literature Review

1. Introduction

Artificial Intelligence (AI) has emerged as a transformative technology in sports management by enabling intelligent, data-driven, and personalized solutions for sports organizations. Traditional sports management systems often depend on manual data collection and analysis for managing players, matches, training activities, and fan engagement. In contrast, AI-powered sports management systems can analyse large volumes of player statistics, match data, training records, and user interactions to provide intelligent insights and recommendations. Technologies such as machine learning, natural language processing (NLP), predictive analytics, computer vision, and recommendation systems play an important role in improving sports performance, decision-making, and fan experiences. Recent developments in generative AI and AI-powered assistants have further expanded the potential of intelligent sports platforms by enabling automated content generation, interactive fan assistance, and personalized services.

2. Review of Existing Literature

Recent research on AI in sports has highlighted the growing use of machine learning and data analytics for analysing athlete performance and supporting sports decision-making. AI-based systems can process player statistics and historical match data to identify performance patterns, evaluate strengths and weaknesses, and provide useful insights for coaches and sports managers. These technologies help reduce dependence on manual analysis and support more informed decisions related to player development, training, and match preparation.

AI has also gained importance in sports fan engagement and communication. AI-powered chatbots and virtual assistants can provide users with quick responses to common queries related to match schedules, players, fixtures, results, and club information. Natural language processing enables these systems to understand user questions and generate relevant responses, improving communication between sports organizations and their supporters.

Another important application of AI in sports management is automated content generation and sports analytics. AI systems can generate match summaries based on match statistics and analyse player performance using available data. Predictive analytics can also be used to identify trends and support decision-making. However, the effectiveness of these systems depends on the quality and availability of reliable sports data. Challenges such as data privacy, system accuracy, explainability, and integration with existing sports management systems remain important considerations.

3. Research Gap

Although AI has been increasingly applied in sports management, several limitations remain in existing systems:

- Many sports platforms focus mainly on individual functions such as player analytics or match prediction rather than providing an integrated management ecosystem.
- Existing football club websites often provide information and news but lack comprehensive club management features.
- Many systems do not integrate AI-powered fan assistance with player performance analysis and club management.
- AI-generated sports content may sometimes produce inaccurate information if it is not based on reliable and verified data.
- There is limited integration of club management, fan engagement, ticket booking, match management, and AI services within a single platform.
- Local football clubs and academies often lack affordable and centralized digital platforms for managing their activities and engaging with supporters.

4. Proposed Research

The proposed research aims to design and develop ClubVerse AI, an intelligent football club management and fan engagement platform that integrates modern web technologies with Artificial Intelligence. The system will provide centralized management of players, coaches, matches, fixtures, training schedules, news, galleries, and ticket bookings. It will also provide an AI Football Assistant that can respond to common user queries related to the club and its activities.

The proposed platform will incorporate AI-based features such as automated match summary generation and player performance analysis. The AI system will analyse available match and player data to generate meaningful insights and assist coaches and administrators in making better decisions. By combining football club management, fan engagement, ticket booking, and AI-powered services into a single platform, ClubVerse AI aims to improve operational efficiency, enhance communication with supporters, and provide a more intelligent and engaging digital experience for football clubs and their communities.

References (APA 7th Edition)

1. Claudino, J. G., Capanema, D. O., de Souza, T. V., Serrão, J. C., Machado Pereira, A. C., & Nassis, G. P. (2019). Current approaches to the use of artificial intelligence for injury risk assessment and performance prediction in team sports: A systematic review. *Sports*

Medicine - Open, 5, 28.

[Read the research paper](#)

2. Rico-González, M., Pino-Ortega, J., & Méndez, A. (2023). Machine learning application in soccer: A systematic review. *Biology of Sport*, 40(1), 249–263.

[Read the research paper](#)

3. Van Eetvelde, H., Mendonça, L. D., Ley, C., Seil, R., & Tischer, T. (2021). Machine learning methods in sport injury prediction and prevention: A systematic review. *Sports Medicine - Open*, 7, 27.

[Read the research paper](#)

4. Vec, V., Tomažič, S., Kos, A., & Umek, A. (2024). Trends in real-time artificial intelligence methods in sports: A systematic review. *Journal of Big Data*, 11, 148.

[Read the research paper](#)

5. Singh, N. (2020). Sport analytics: A review. *The International Technology Management Review*.

[Read the research paper](#)

6. Kranzinger, S., Halmich, C., Hofer, D., et al. (2026). A scoping review of explainable artificial intelligence in sports science. *Discover Artificial Intelligence*, 6, 5.

[Read the research paper](#)